

# Integrated Dataset Generation and Testing Platform for Dnynamic Graph Neural Network ETA Prediction

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## Abstract

Accurate Estimated Time of Arrival (ETA) prediction is essential for navigation, traffic management, and smart transportation. Dynamic Graph Neural Networks (GNNs) model evolving road networks, but research is hindered by limited datasets and the lack of a unified environment for data generation and model testing.

We present an integrated open-source platform combining **Traffic-DSTG-Gen** (route-aware dynamic spatio-temporal graphs from SUMO simulation or GPS trajectories) and **TrafficLab** (interactive web-based ETA evaluation). The system supports 80,000+ simulation snapshots with 28-dimensional node and 7-dimensional edge features, and logs thousands of real-time test journeys with MAE/RMSE analysis.

## Keywords

Graph Neural Networks; ETA Prediction; SUMO Simulation; Dataset Generation; Route-Aware Graphs; Testing Platform

## II. Principle

ETA prediction requires data that links road topology, vehicle motion, and trip labels. Static-sensor and plain trajectory datasets often lack explicit routes and rich features for dynamic GNNs.

**Principle:** one shared *route-conditioned dynamic graph*, built via (1) **simulation**—SUMO/TraCI with reproducible snapshots; or (2) **trajectory conversion**—GPS traces map-matched to the same schema (e.g. Porto taxi). Both paths feed one training pipeline and one web evaluation client.

## III. Theoretical Modelling

**Hybrid graph:** static junction nodes and road edges; dynamic vehicle nodes; time-varying junction–vehicle–vehicle–junction edges. **Features:** 28-D nodes, 7-D edges; sequential snapshots for temporal models.

**Traffic-DSTG-Gen** (desktop, PySide6 + SUMO): scenario configuration, trajectory import, export. **TrafficLab** (Vue.js + FastAPI + PostgreSQL): route selection, ETA inference, journey persistence, cohort metrics.

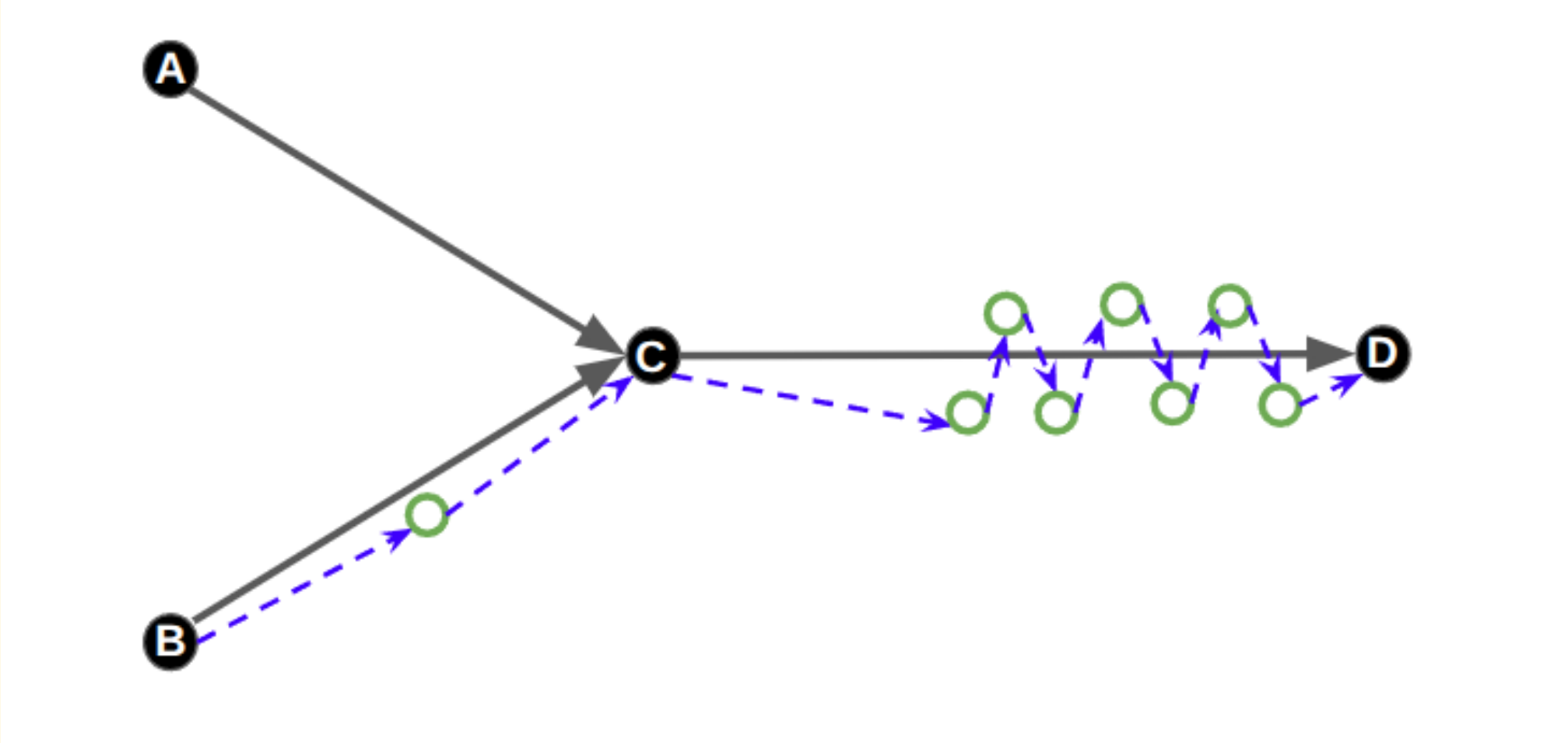


Fig. 1. Static and dynamic graph structure: junction nodes (black), vehicle nodes (green), static roads (grey), dynamic relations (blue dashed).

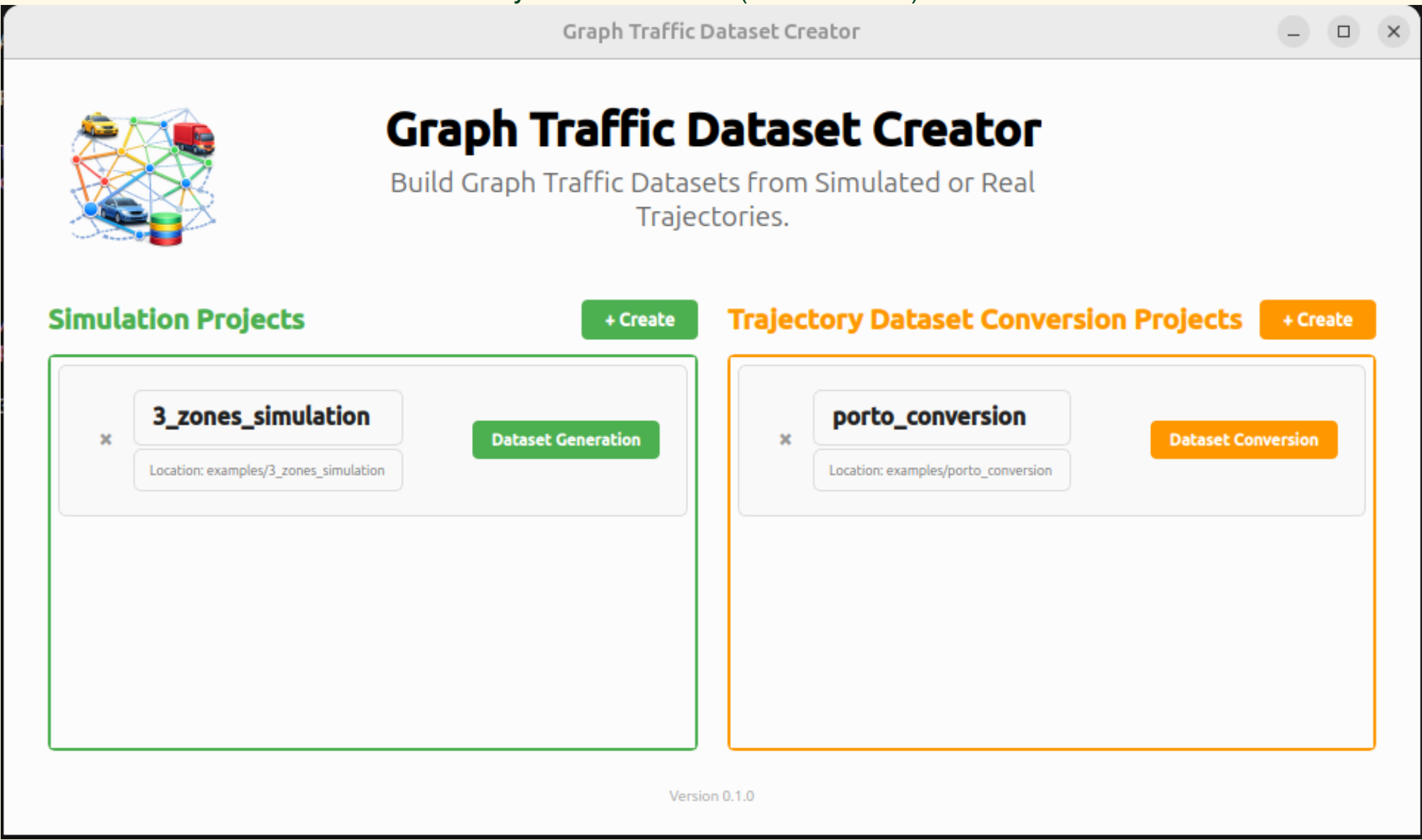


Fig. 2. Traffic-DSTG-Gen desktop: 3-zone city simulation and Porto taxi conversion projects.

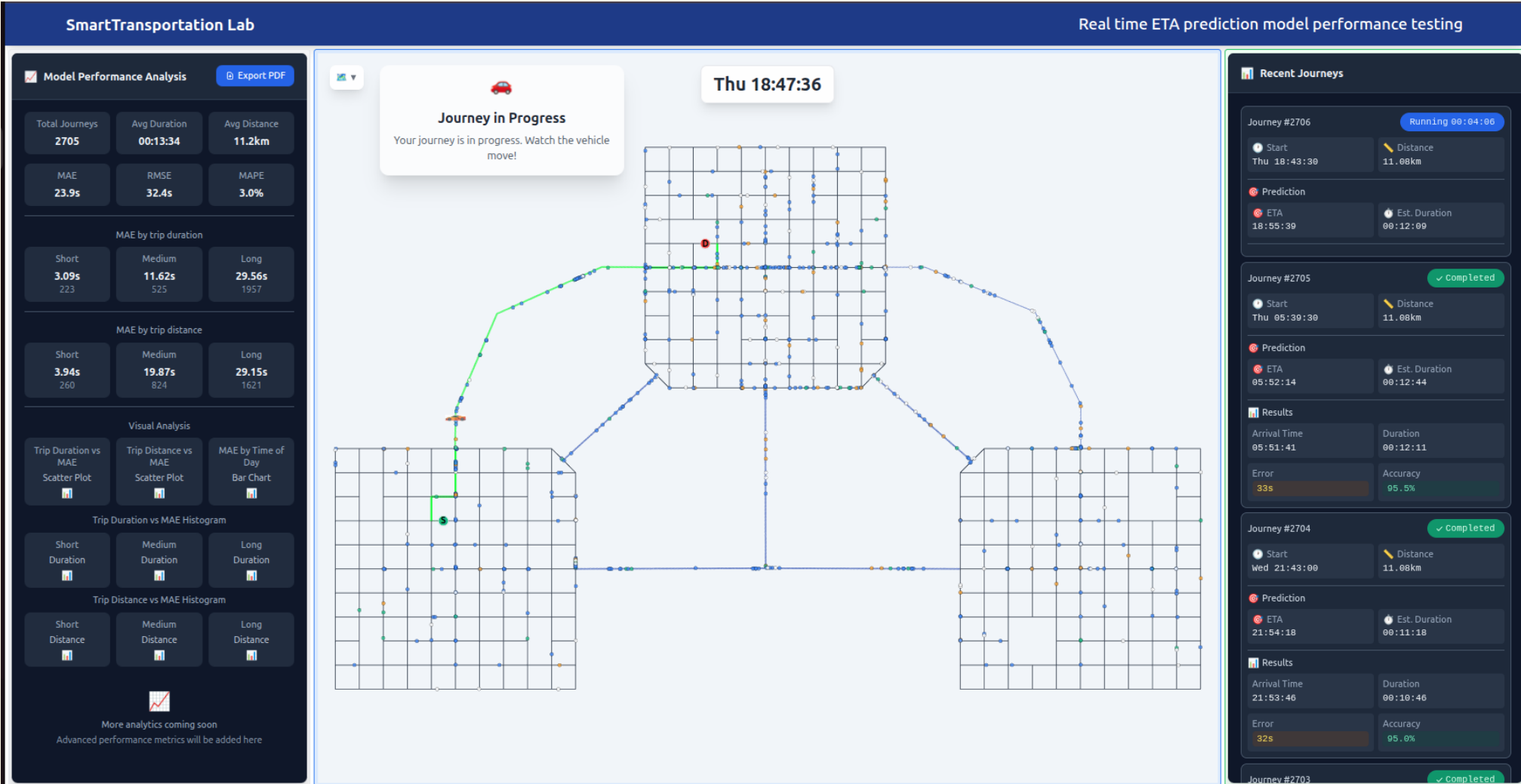


Fig. 3. TrafficLab web UI: interactive ETA testing on a SUMO network.

## V. Results and Analyses

**Dataset scale:** 80,000+ snapshots; multi-zone urban scenarios. **Web evaluation:**

2,707 simulation and 3,055 trajectory journeys logged independently of training data.

Table I. Web-tool ETA error (seconds) vs. held-out training evaluation.

| Source     | <i>n</i> | MAE  | RMSE  | MedAE |
|------------|----------|------|-------|-------|
| Simulation | 2,707    | 46.2 | 107.2 | 31.2  |
| Trajectory | 3,055    | 52.4 | 121.5 | 35.5  |

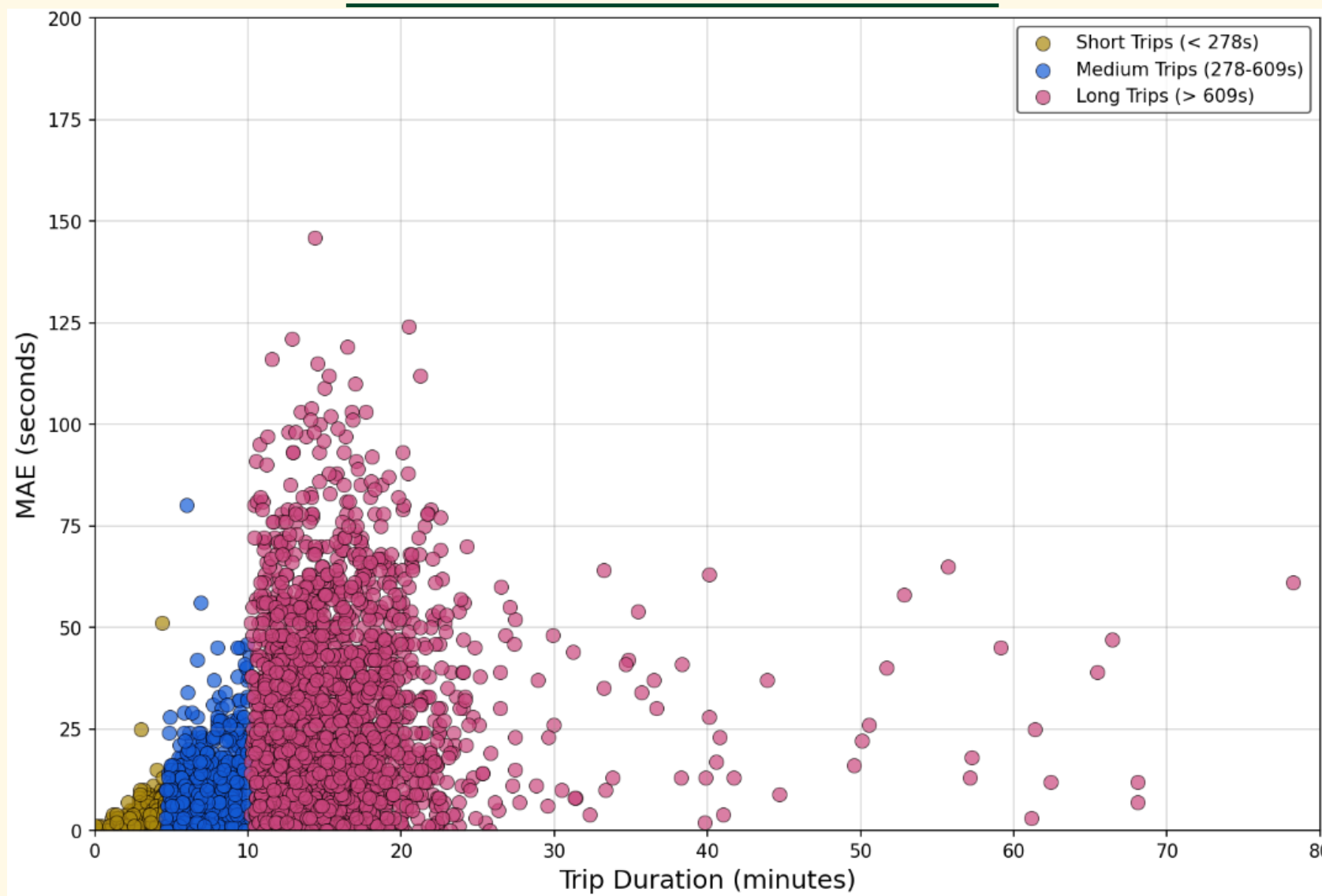


Fig. 4. MAE vs. trip duration; per-journey errors align with training evaluation within ~2s MAE.

## VI. Conclusion

Integrated platform for dynamic GNN ETA research: dataset generation (simulation or trajectories) and real-time testing in one workflow. Open source: <https://github.com/Ruppin-SmartTransportation/Traffic-DSTG-Gen> and TrafficLab; demo: <https://demo.smart-transport.cloud/>.

**Future work:** federated learning, additional data sources, larger-network deployment.

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